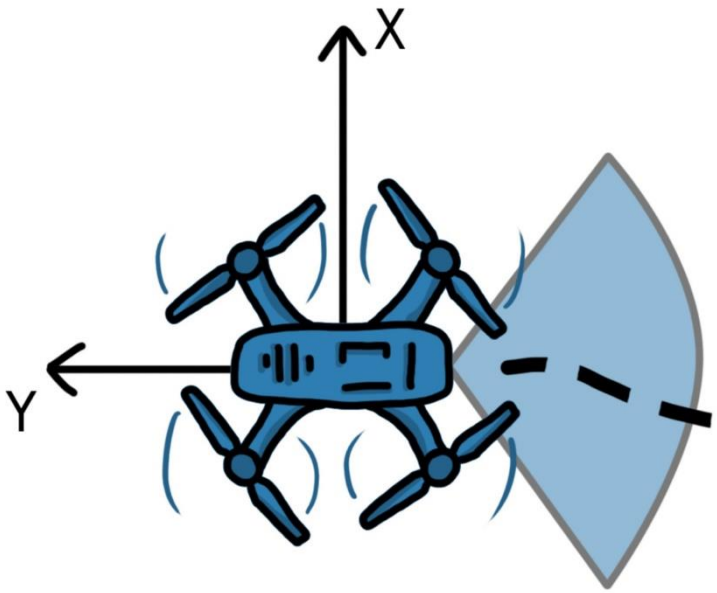
小迈步之人工智能（四）：

实战传感器融合与目标跟踪



王鸿钧，高级应用工程师
MathWorks 中国

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多旋翼无人机的硬件支持包资源



Ryze Tello Drone Support from MATLAB



PX4 Autopilots Support from Embedded Coder



Parrot Minidrones Support from Simulink

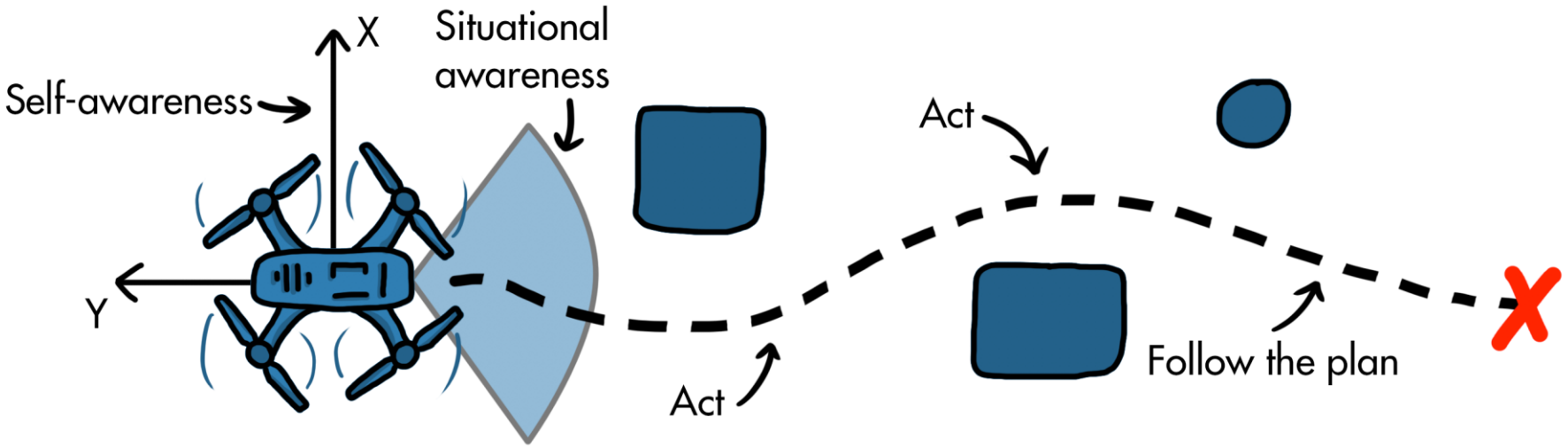


AR.Drone 2.0 Support from Embedded Coder



多旋翼无人机——自主系统如何产生自主能力?

自主系统的目标是在不与人类交互的环境中运行。系统需要能够理解自身和它周围的世界，然后决定采取哪条路径和哪些命令是正确的，并使系统跟随这条路径。



我们可以把这些自主能力分为四个主要领域：
传感 (Sense)、感知 (Perceive)、决策与规划 (Decide & Plan) 和行动 (Act)。



自主系统的四种通用能力

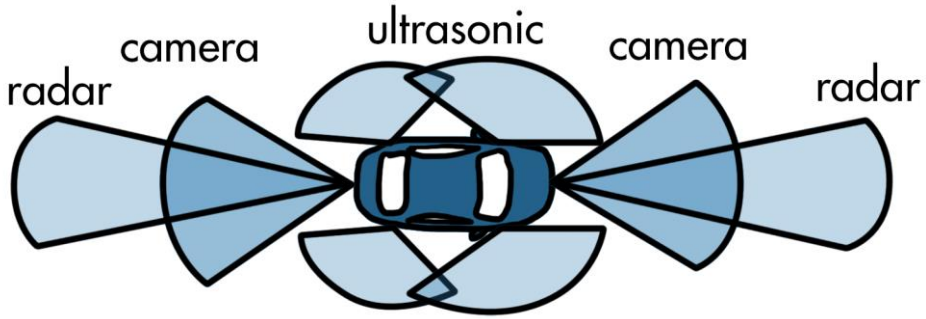
传感（Sense）

传感，是指使用传感器收集有关系统状态和外部世界状态的信息。例如，一个自动驾驶系统可以测量它自身的状态——使用 GPS，来确定它的位置；同时它也可以测量环境的状态，使用诸如雷达、激光雷达和视觉相机等面向外部的传感器。

选定的传感器组合依赖于具体应用，但这一步的目标始终是相同的：收集足够的信息，使系统能够正确地领会所处状况，并最终采取行动。

当然，简单地收集到的原始传感器数据并不能立即发挥作用。系统还必须理解这些数据。这就是下一步的任务——感知。

自动驾驶系统的传感器组合示例



感知（Perceive）

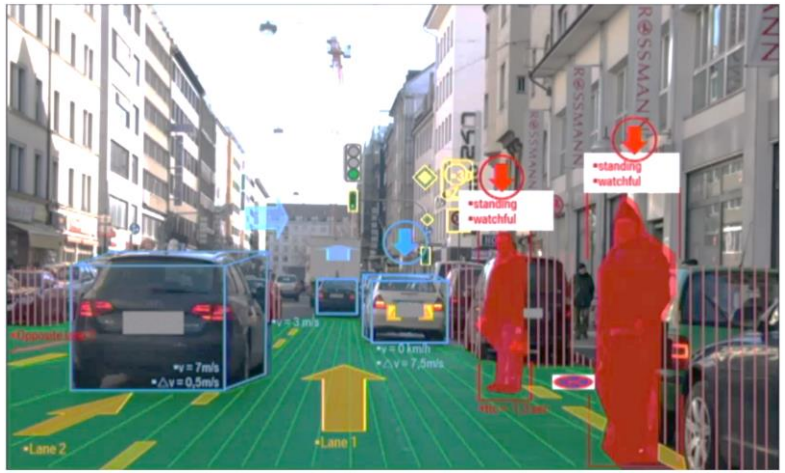
理解感知的最简单方法，可以想想相机传感器的原始图像。每帧图像可能有数百万像素，每个像素包含三个不同色彩的8位或更多信息。这是巨大的数据量。为了利用这个庞大的数组，需要一个算法或者一个人来找出一些有价值的东西。也就是说，感知不仅仅是获取信息，还要把它解释成一个有用的量。

例如，左边的图像是相机的原始输出。自动驾驶汽车最终必须将这些像素点解释为带有车道线的道路，并识别出有其他车辆和行人正在共享该空间。它还必须识别交通标志，从而获得路况的附加信息。

传感



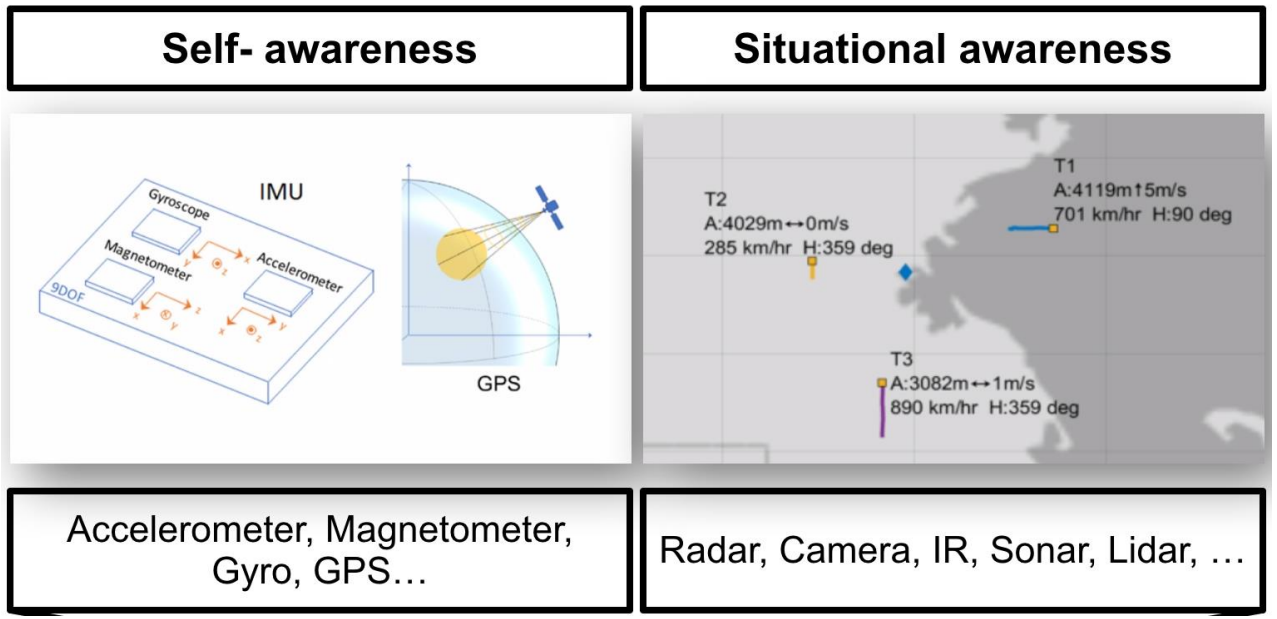
感知



感知（Perceive）

感知有两个不同但同等重要的责任。它负责“自我意识”，也就是感知自身的状态；以及“情境意识”，也就是感知环境中的其他物体并跟踪它们。

我在哪里？
我在做什么？



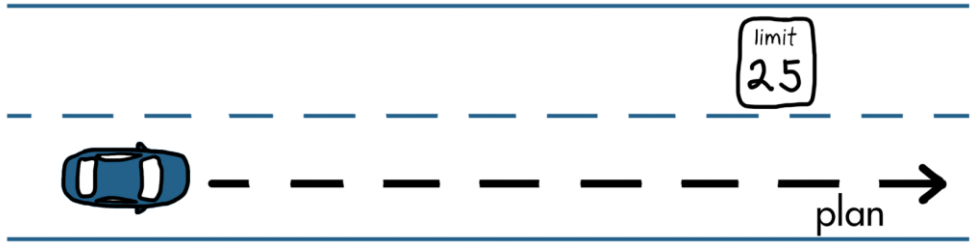
那是什么？
它在做什么？

一个系统需要能够理解它自身和它周围的环境，以便有足够的信息来做出决策。

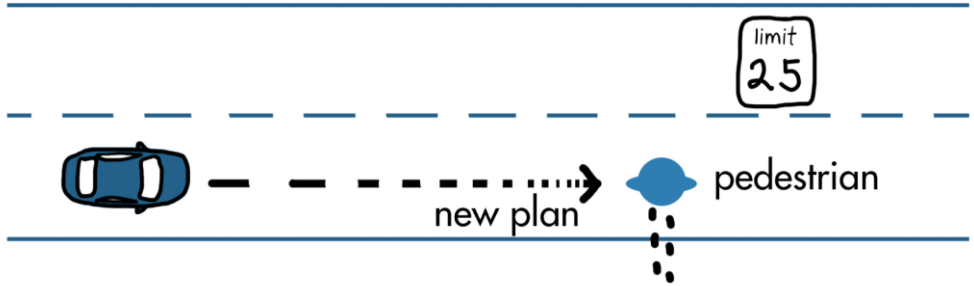
决策与规划（Decide & Plan）

决策与规划，是指自主系统决定要做什么，并制定到达目标的路径。

例如，让我们看看车辆的巡航控制——一种相对简单的自主行为。巡航控制的目标是，即使存在扰动，也要保持一个固定的速度。在这里，巡航控制的规划被设置为道路的限速。



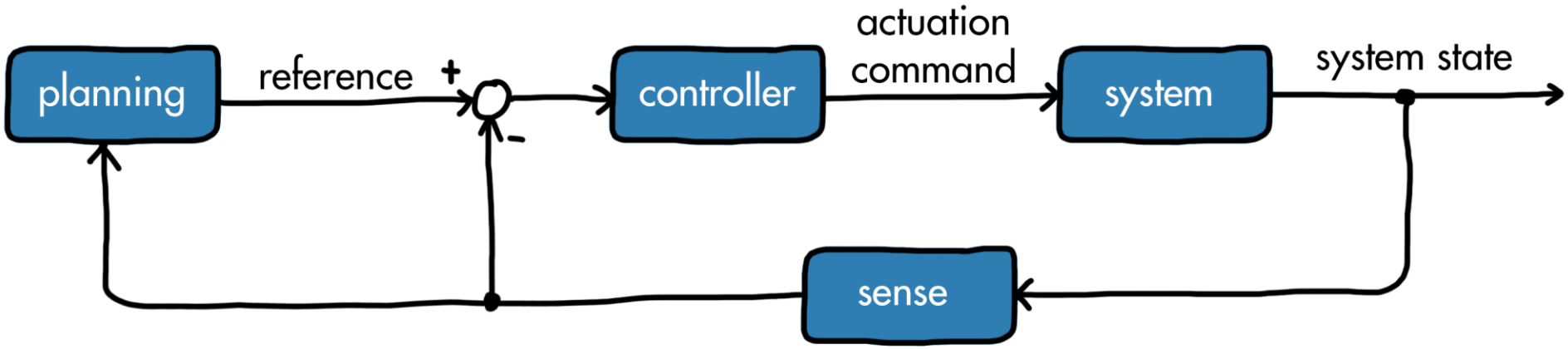
然而，如果车流速度低于限速，或者有行人走到车前，这个规划就会改变。车辆通过传感了解环境的变化，通过感知获得障碍物信息，制定一个新的规划来减速或停车，然后控制车辆跟随这个规划。



行动（Act）

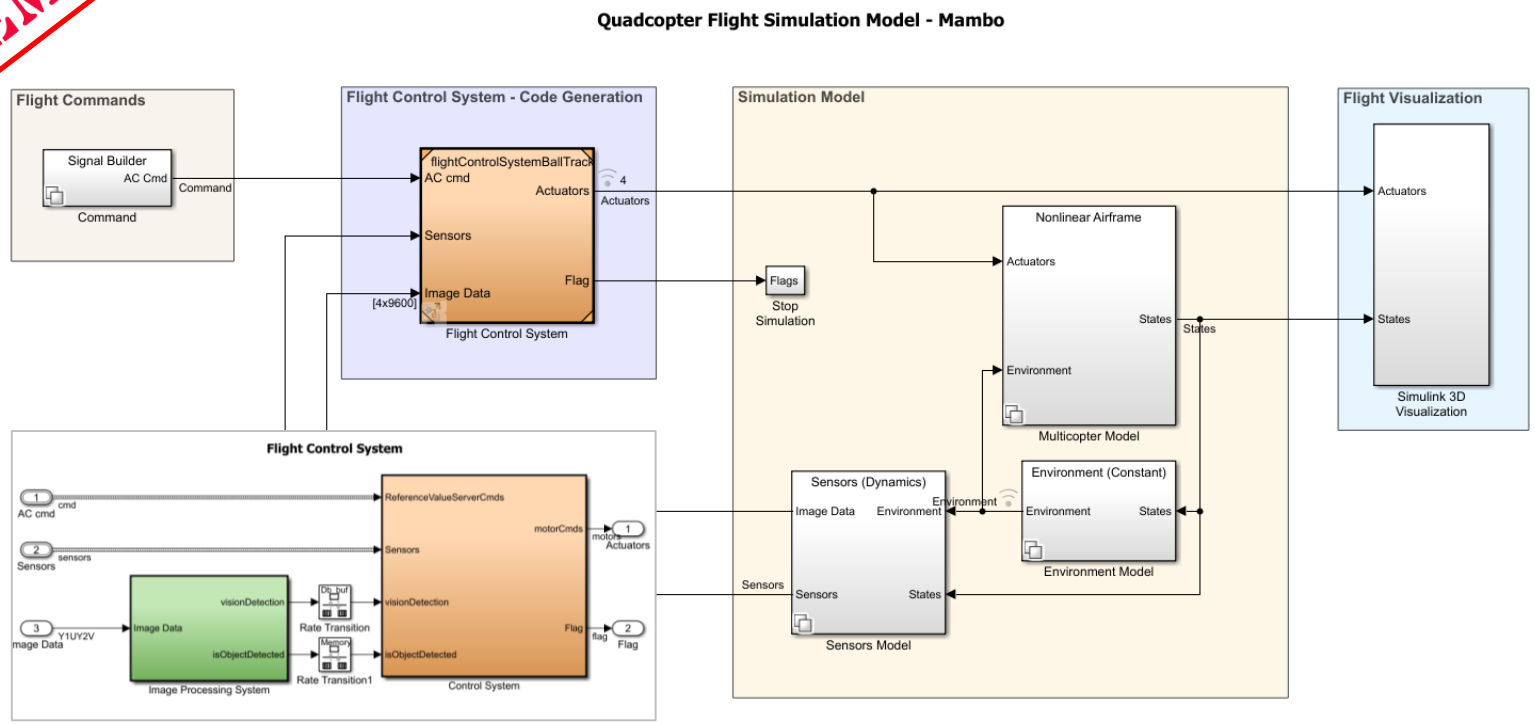
自主系统的最后一步：计算和执行反馈控制，使系统跟随规划。

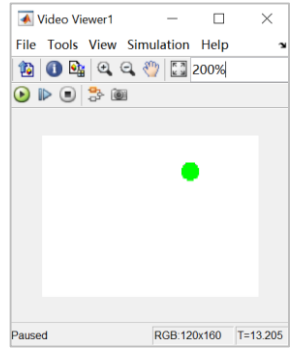
我们可以把规划看作提供一个参考信号，还要有控制器来控制执行器和其他机构，操作系统沿规划的路径移动。在车辆巡航的示例中，这一行为由巡航控制器完成。该控制器可以感知实际车速，并将其与参考车速进行比较，然后控制发动机产生必要的动力，将误差降至零。

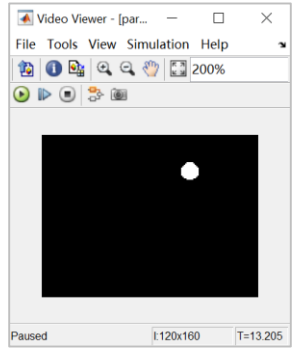


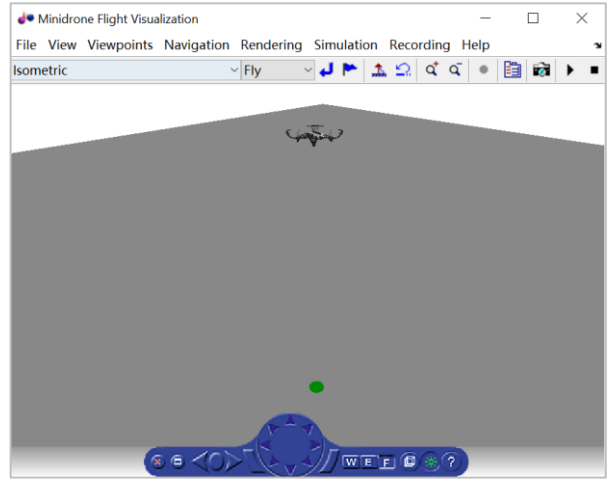
使用无人机实现简单的自主任务——小球跟踪

DEMO





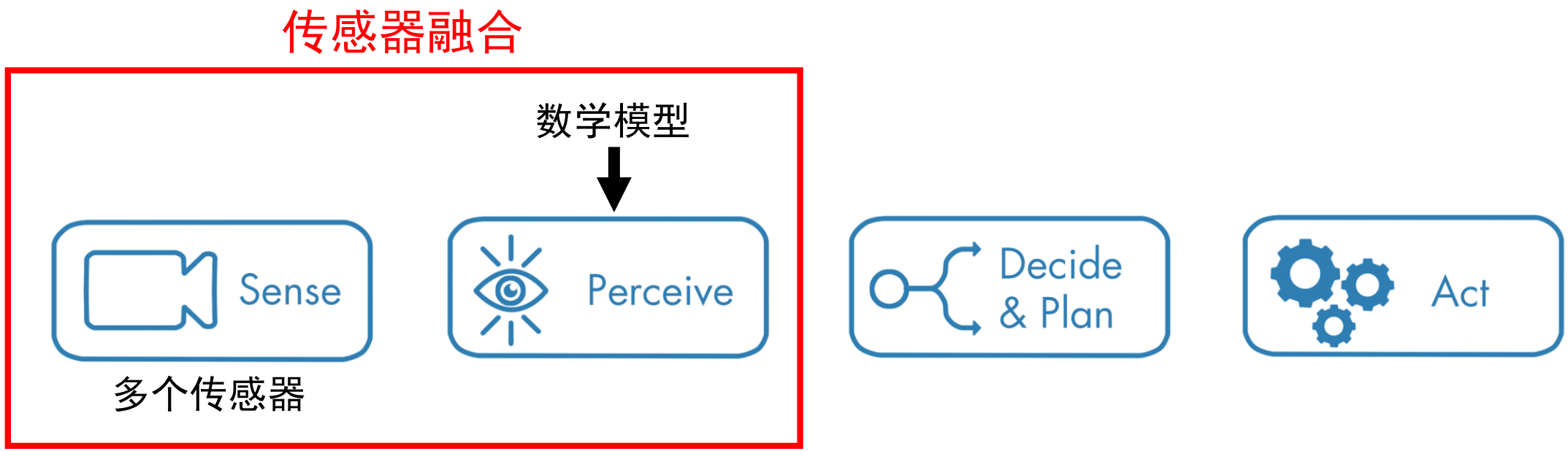




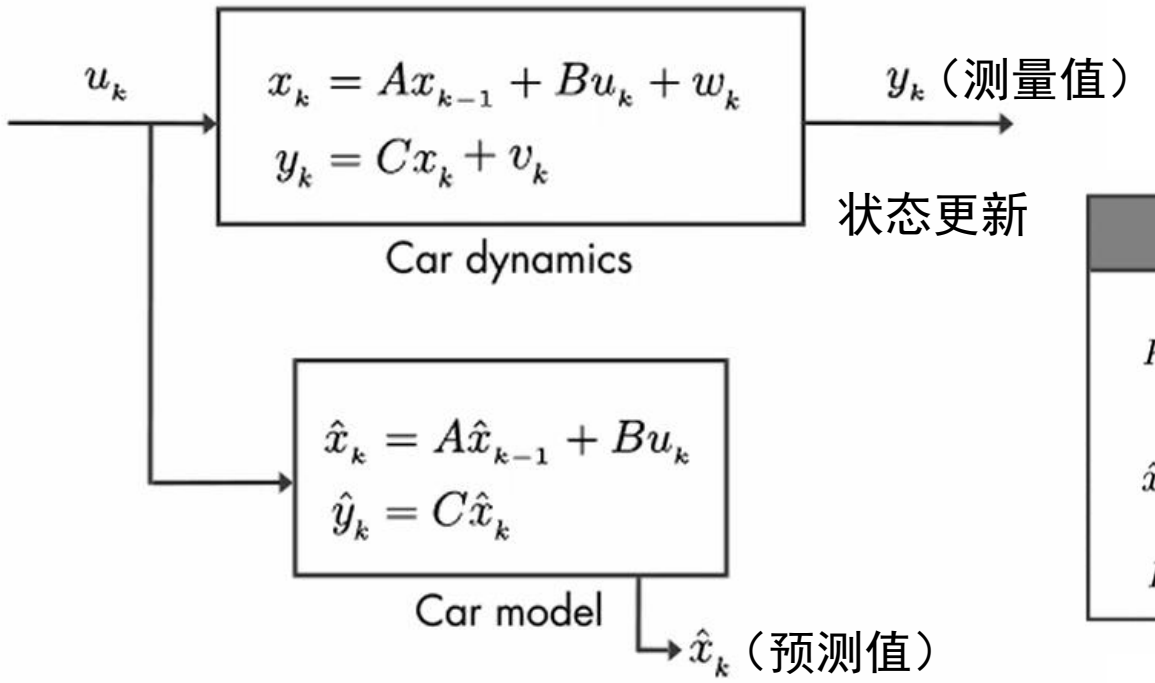
实战传感器融合——定位篇

什么是传感器融合？

传感器融合涉及传感和感知两个方面。传感器融合是一种方法，我们可以混合来自多个传感器的测量数据，并将它们与来自数学模型的附加信息结合起来，从而更好地理解世界。



使用什么样的数学模型？



状态更新

状态预测

卡尔曼滤波（Kalman Filter）

是一个数学迭代过程，使用一组方程，结合包含噪声的测量值和来自过程模型的预测值，实现对真实值的最优估计。

Update

$$K_k = \frac{P_k^- C^T}{CP_k^- C^T + R}$$

$$\hat{x}_k = \hat{x}_k^- + K_k(y_k - C\hat{x}_k^-)$$

$$P_k = (I - K_k C)P_k^-$$

Prediction

$$\hat{x}_k^- = A\hat{x}_{k-1} + Bu_k$$

$$P_k^- = AP_{k-1}A^T + Q$$

如何实现卡尔曼滤波？

通过 Control System Toolbox:

```
Plant = ss(A,[B B],C,0,-1,'inputname',{'u' 'w'},'outputname','y');
[kalmf,L,P,M] = kalman(Plant,Q,R);
```

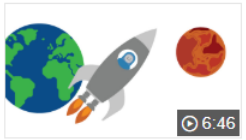


请参考：https://ww2.mathworks.cn/campaigns/offers/kalman-filter-design-examples.html

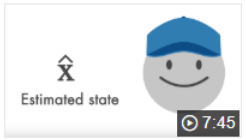
通过 Computer Vision Toolbox:

```
kalmf = vision.KalmanFilter(stateModel,measurementModel,'ProcessNoise',Q,'MeasurementNoise',R);
```

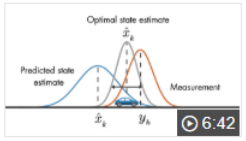
Melda Ulusoy的卡尔曼滤波系列视频讲座



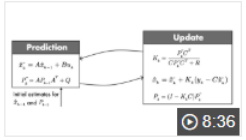
Part 1: Why Use Kalman Filters?
Discover common uses of Kalman filters by walking through some examples. A Kalman filter is an optimal estimation algorithm used to estimate states of a system from indirect and uncertain measurements.



Part 2: State Observers
Learn the working principles of state observers, and discover the math behind them. State observers are used to estimate the internal states of a system when you can’ t directly measure them.



Part 3: An Optimal State Estimator
Learn how Kalman filters work. Kalman filters combine two sources of information, the predicted states and noisy measurements, to produce optimal, unbiased state estimates.



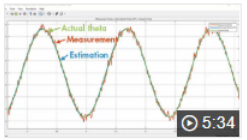
Part 4: An Optimal State Estimator Algorithm
Discover the set of equations you need to implement the Kalman filter algorithm.



Part 5: Nonlinear State Estimators
This video explains the basic concepts behind nonlinear state estimators, including extended Kalman filters, unscented Kalman filters, and particle filters.



Part 6: How to Use a Kalman Filter in Simulink
Estimate the angular position of a simple pendulum system using a Kalman filter in Simulink. You will learn how to configure Kalman filter block parameters such as the system model, initial state estimates, and noise characteristics.



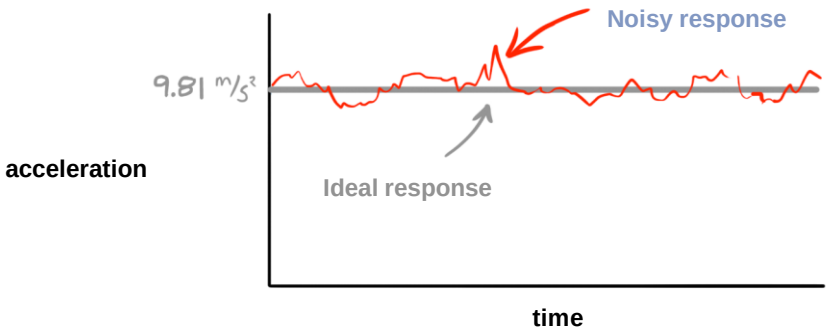
Part 7: How to Use an Extended Kalman Filter in Simulink
Estimate the angular position of a nonlinear pendulum system using an extended Kalman filter. You will learn how to specify Extended Kalman Filter block parameters such as state transition and measurement functions, and generate C/C++ code.

https://ww2.mathworks.cn/videos/series/understanding-kalman-filters.html

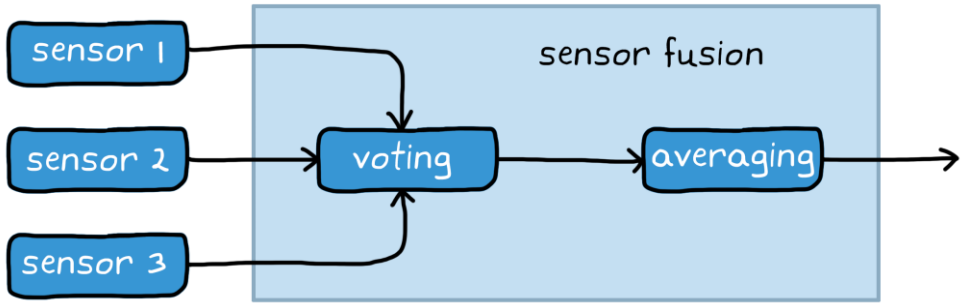
传感器融合有什么好处？

我们可以融合多个数据源，从而：

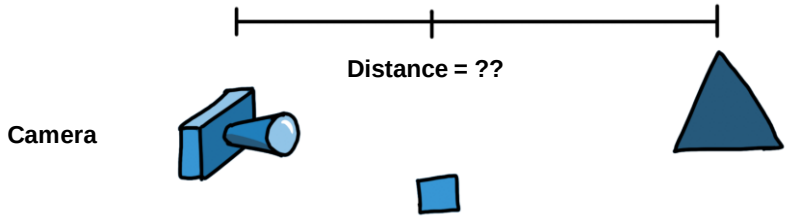
提高数据质量



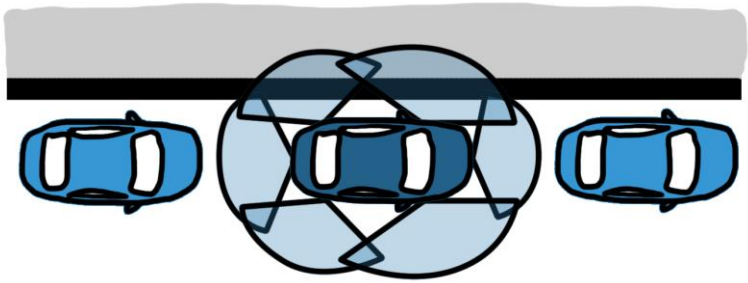
提高数据可用性



估计未测量状态



扩大覆盖区域



更多详细内容：Sensor Fusion and Tracking for Autonomous Systems: An Overview

Brian Douglas的传感器融合系列视频讲座

Series: Understanding Sensor Fusion and Tracking



Part 1: What is Sensor Fusion?
This video provides an overview of what sensor fusion is and how it helps in the design of autonomous systems. It also covers a few scenarios that illustrate the various ways in which sensor fusion can be implemented.



Part 2: Fusing a Mag, Accel, and Gyro to Estimate Orientation
This video describes how we can use a magnetometer, accelerometer, and a gyro to estimate an object's orientation. The goal is to show how these sensors contribute to the solution, and to explain a few things to watch out for along the way.



Part 3: Fusing a GPS and IMU to Estimate Pose
This video describes how we can use a GPS and an IMU to estimate an object's orientation and position. We'll go over the structure of the algorithm and show you how the GPS and IMU both contribute to the final solution.



Part 4: Tracking a Single Object With an IMM Filter
This video describes how we can track a single object by estimating state with an interacting multiple model filter. We build up some intuition about the IMM filter and show how it is a better tracking algorithm than a single model Kalman filter.



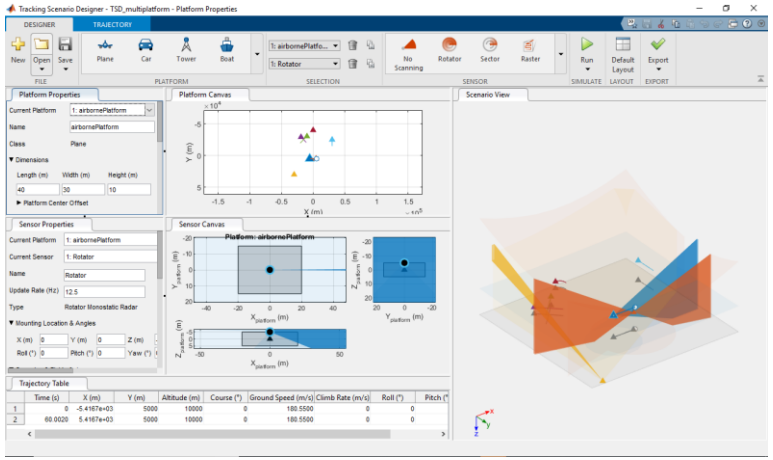
Part 5: How to Track Multiple Objects at Once
This video describes two common problems that arise when tracking multiple objects: data association and track maintenance. We cover a few ways to solve these issues and provide a general way to approach all multi-object tracking problems.

https://ww2.mathworks.cn/videos/series/understanding-sensor-fusion-and-tracking.html

Sensor Fusion and Tracking Toolbox
传感器融合与跟踪工具箱

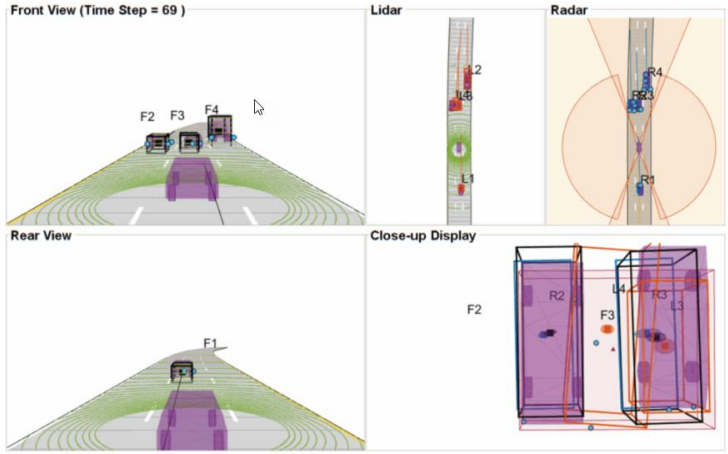
发布于 R2018b

实用工具



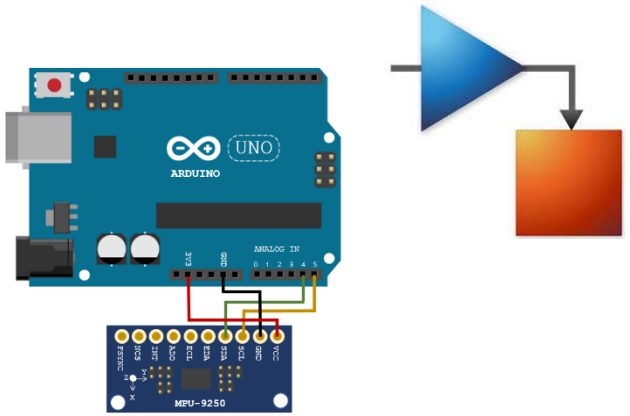
跟踪场景设计器

参考应用



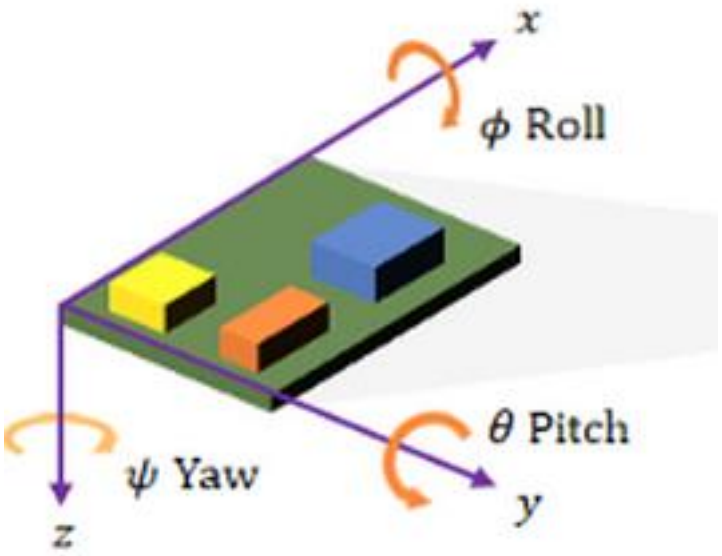
IMU/GPS/视觉/激光雷达/毫米波雷达 …

硬件连接

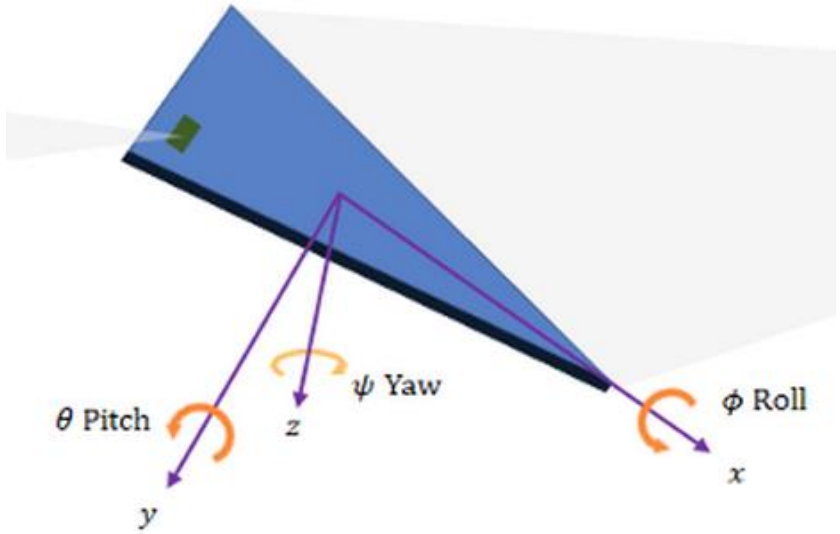


与ARDUINO等硬件通信

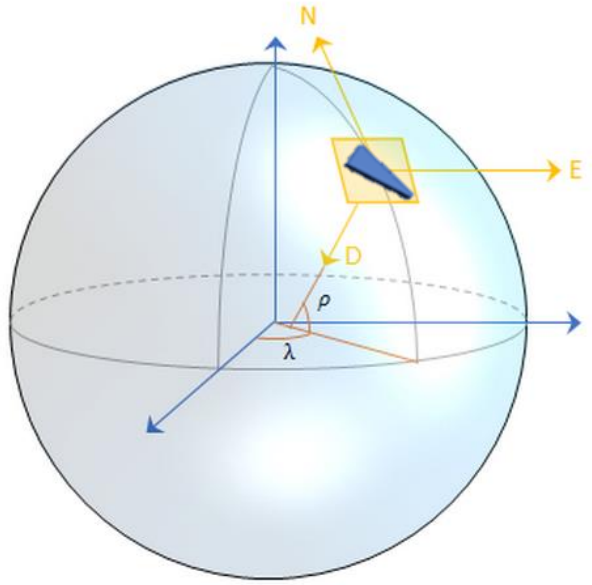
如何实现无人机的位姿估计？



惯性传感器



姿态估计



位置估计

3轴加速度计

3轴陀螺仪



光流传感器（x, y轴定位）

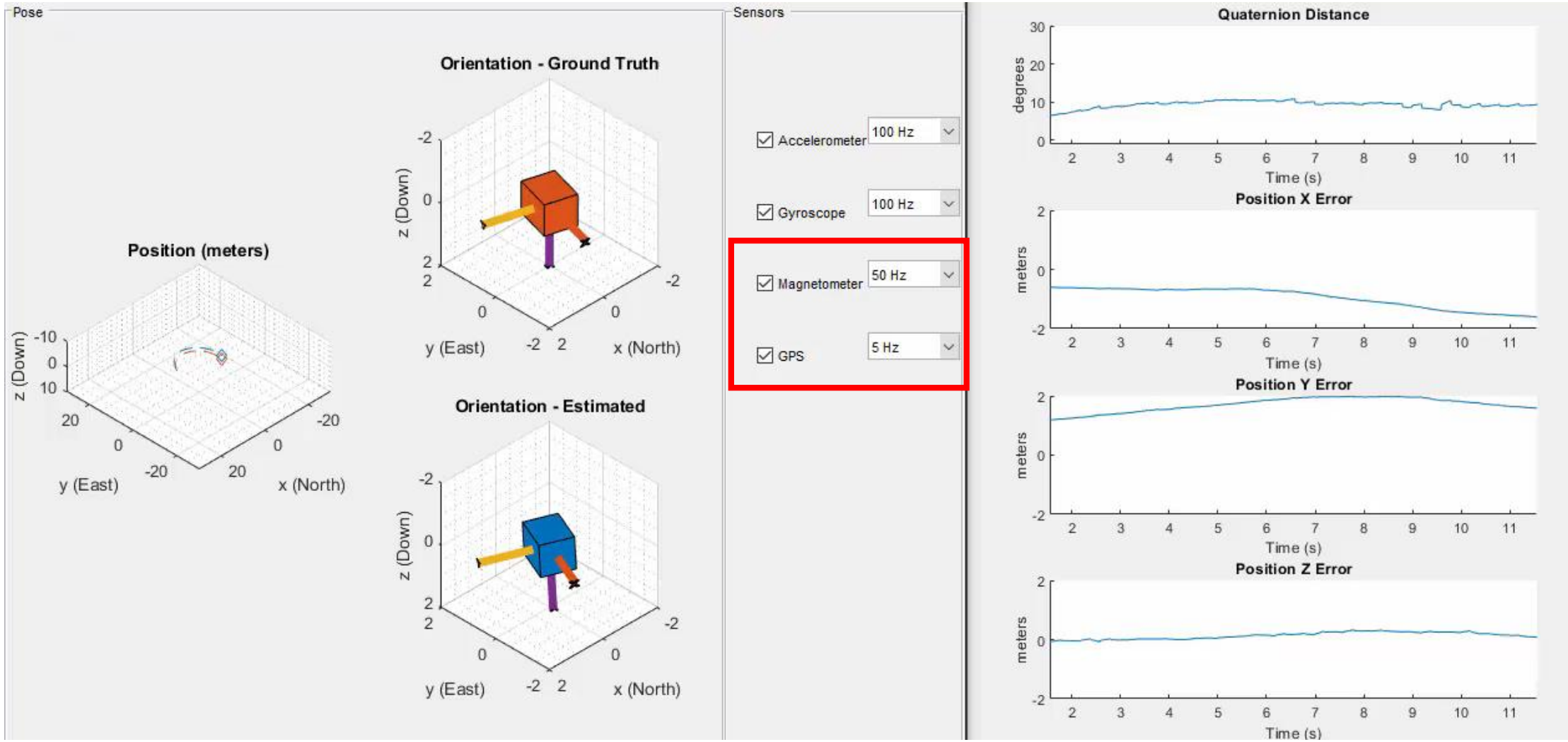
气压计，超声波传感器（z轴定位）

- 为什么不使用GPS？
- 不适合室内定位。

用惯性传感器定位的示例

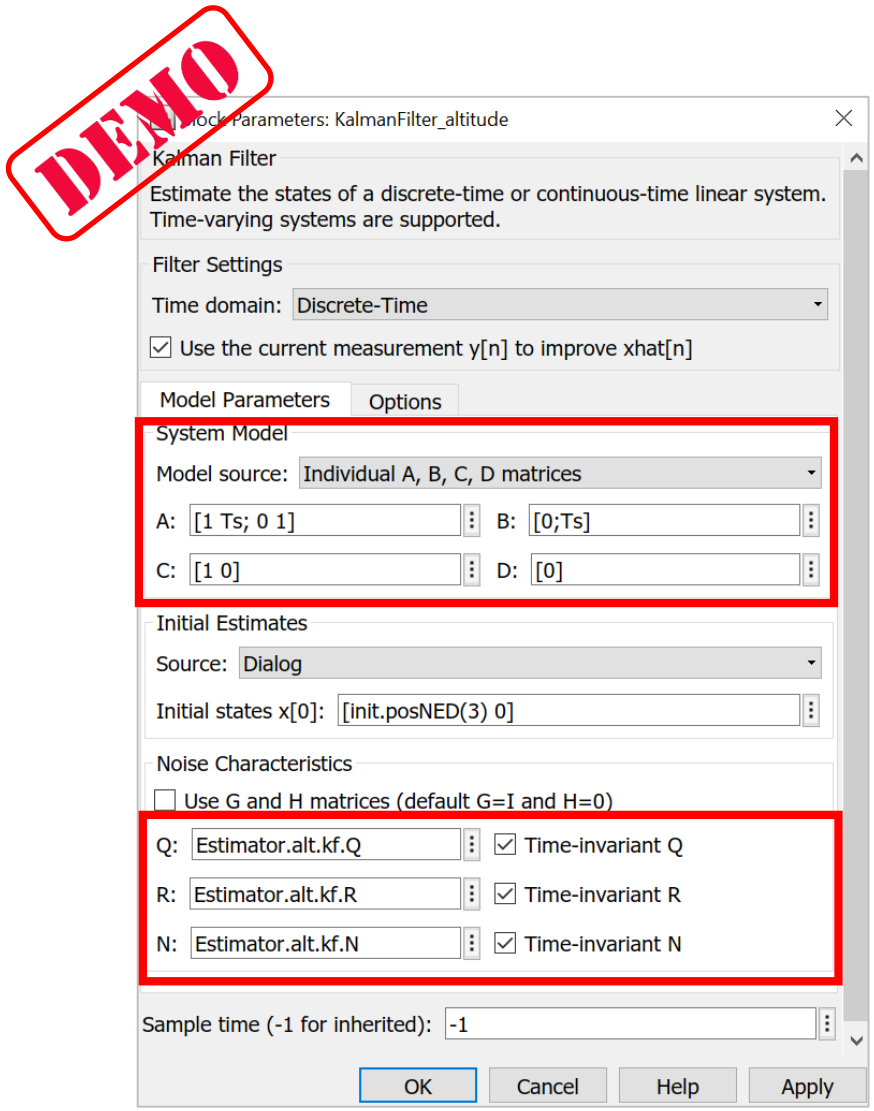
```
fusionfilt = insfilterAsync('ReferenceLocation', refloc);
```

真值 vs. 估计值　　传感器　　估计误差




```
» openExample('PoseEstimationFromAsynchronousSensorsExample')
```

实现无人机的定位和卡尔曼滤波调参



高度（z轴）的位置估计

过程的数学模型：

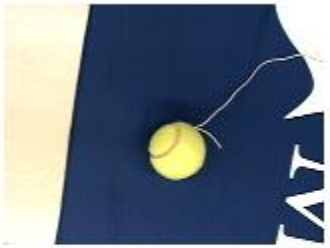
$$\begin{pmatrix} z(k+1) \\ v_z(k+1) \end{pmatrix} = \begin{pmatrix} 1 & T_s \\ 0 & 1 \end{pmatrix} \begin{pmatrix} z(k) \\ v_z(k) \end{pmatrix} + \begin{pmatrix} 0 \\ T_s \end{pmatrix} a_z(k)$$

过程噪声协方差Q：这个值越小，越信任数学模型的预测；
测量噪声协方差R：这个值越小，越信任传感器的测量；

实战传感器融合——障碍物跟踪篇

检测绿色小球的图像算法







```
frame = imread('rgb.bmp');
imshow(frame)

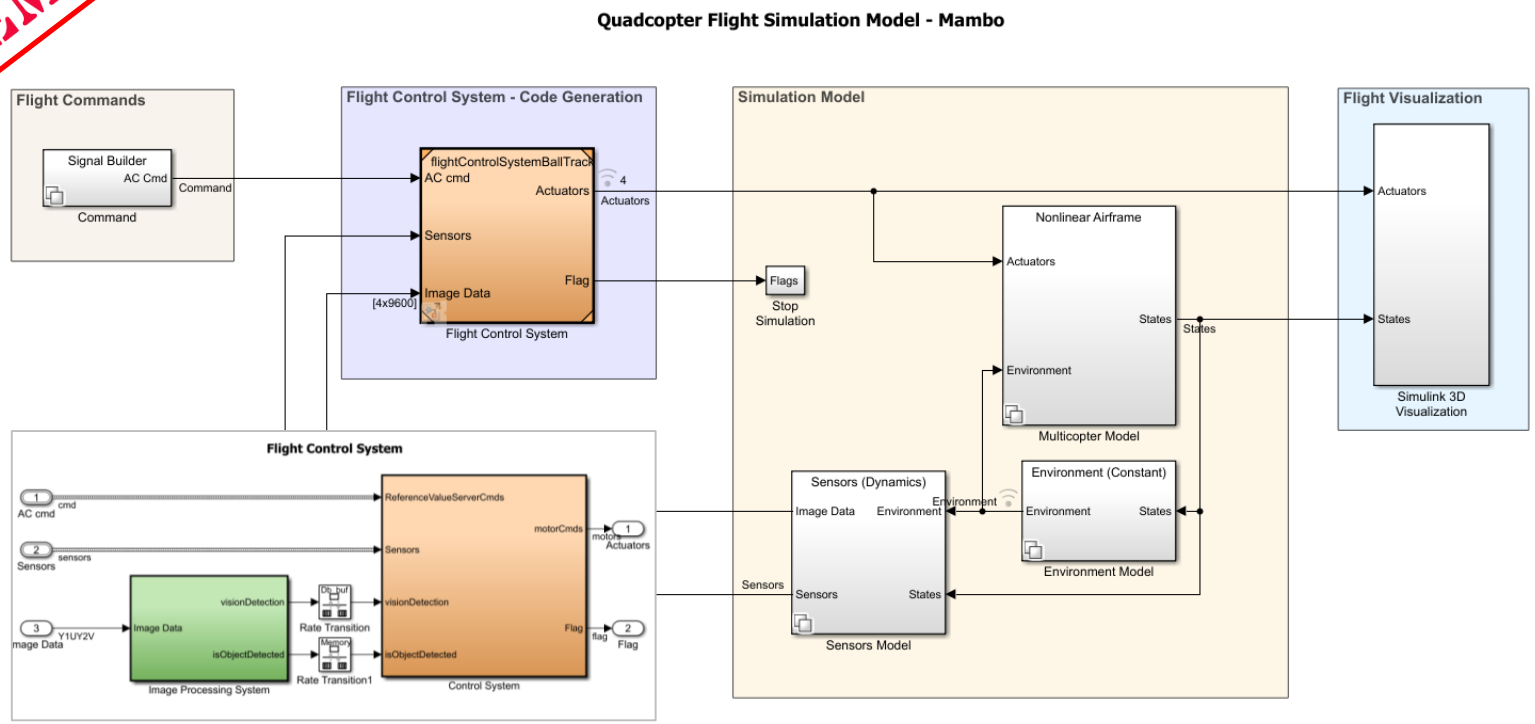
r = frame(:,:,1);
g = frame(:,:,2);
b = frame(:,:,3);
grayImage = g - r/2 - b/2;

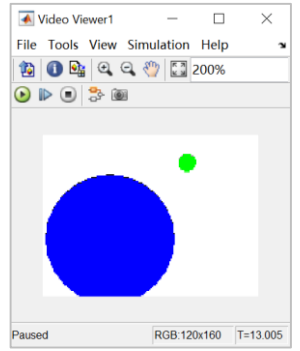
bw = grayImage > 40;
imshow(bw)

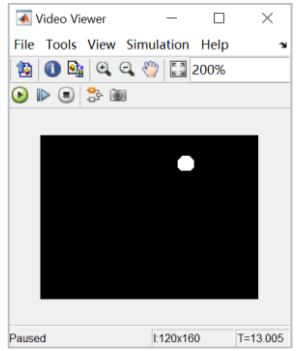
[x, y] = find(bw);
isDetected = size(x,1) > 20;
if isDetected
    xm = round(mean(x));
    ym = round(mean(y));
end
```

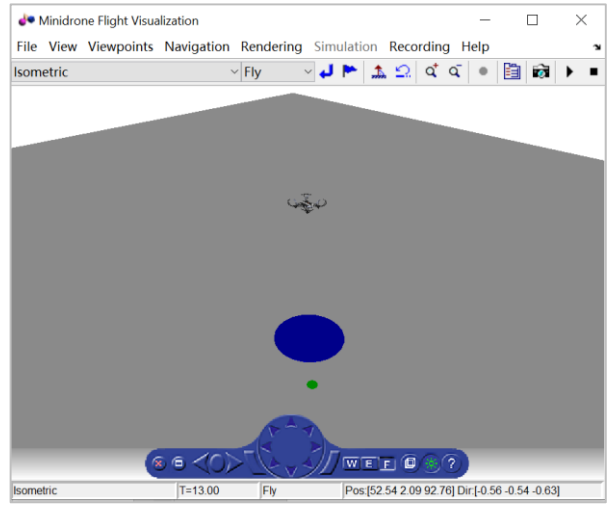
通过卡尔曼滤波跟踪绿色小球

DEMO









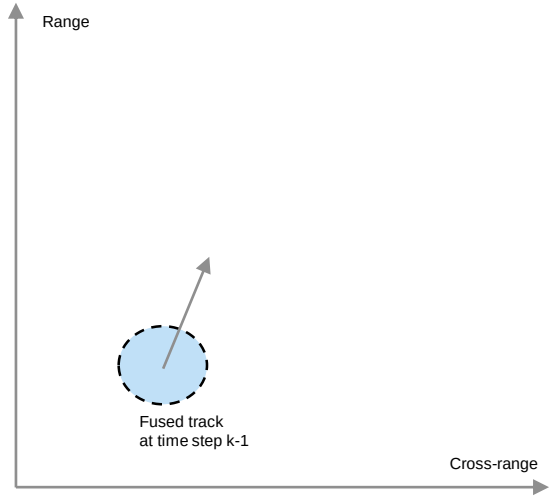
无人机实机跟踪绿色小球



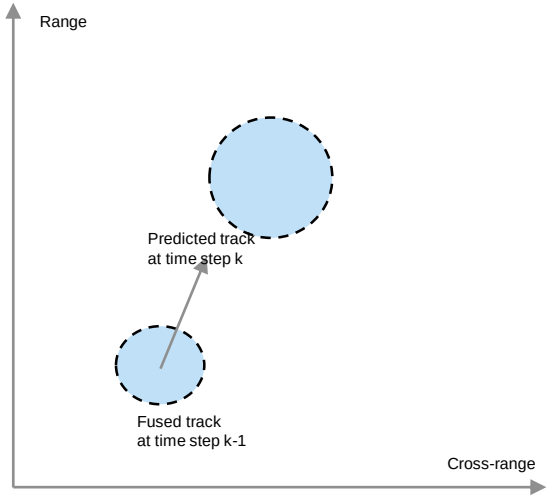


Q&A

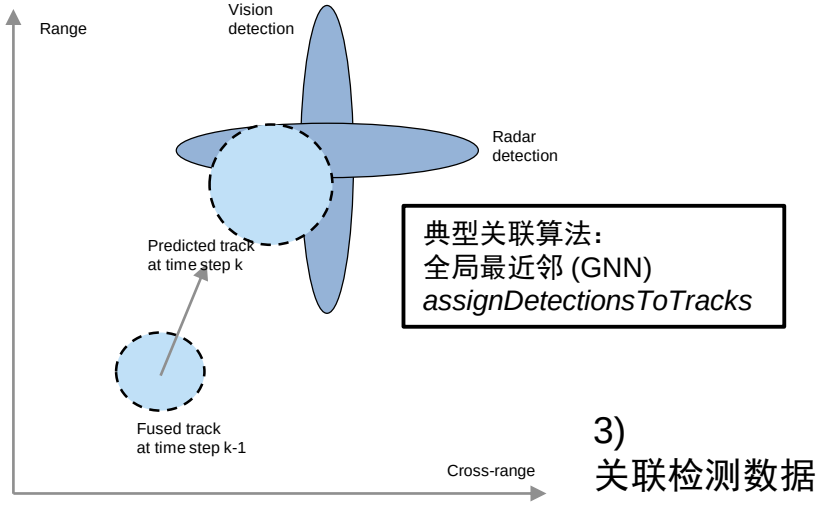
基于多目标跟踪的传感器融合



1)
前一时刻状态

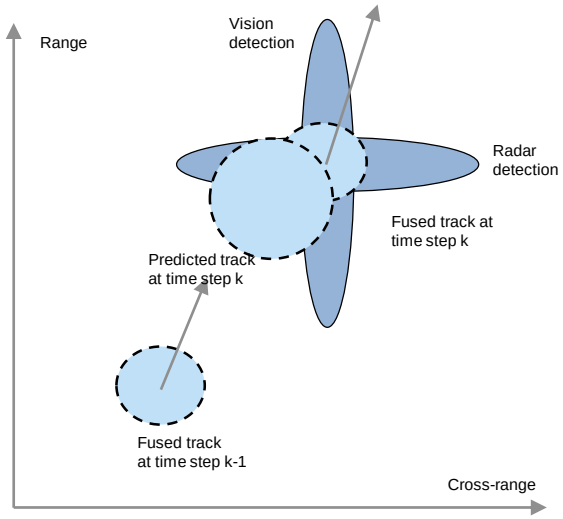


2)
当前时刻预测



典型关联算法:
全局最近邻 (GNN)
assignDetectionsToTracks

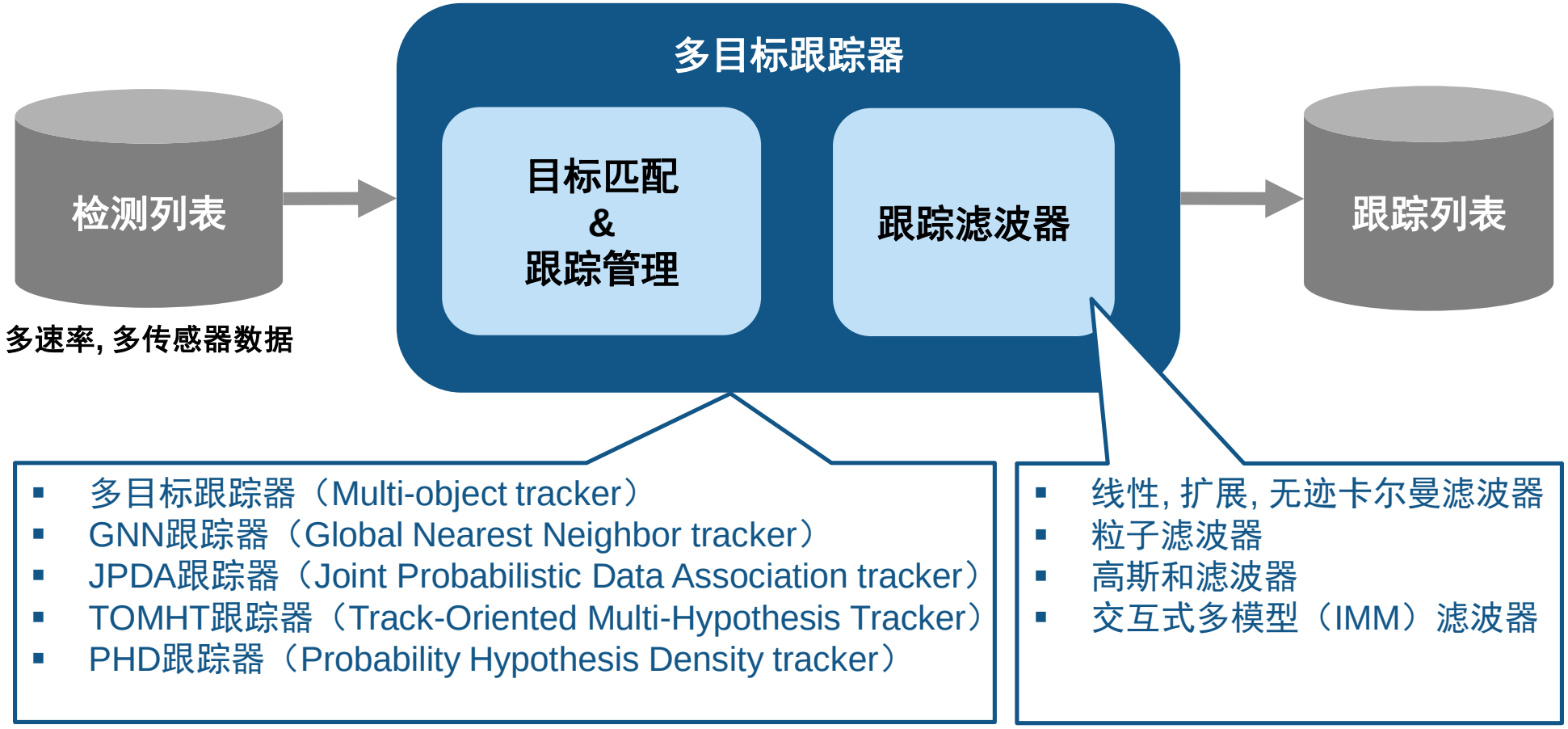
3)
关联检测数据



4)
更新当前状态

传感器融合与跟踪工具箱

自定义跟踪器与滤波器算法

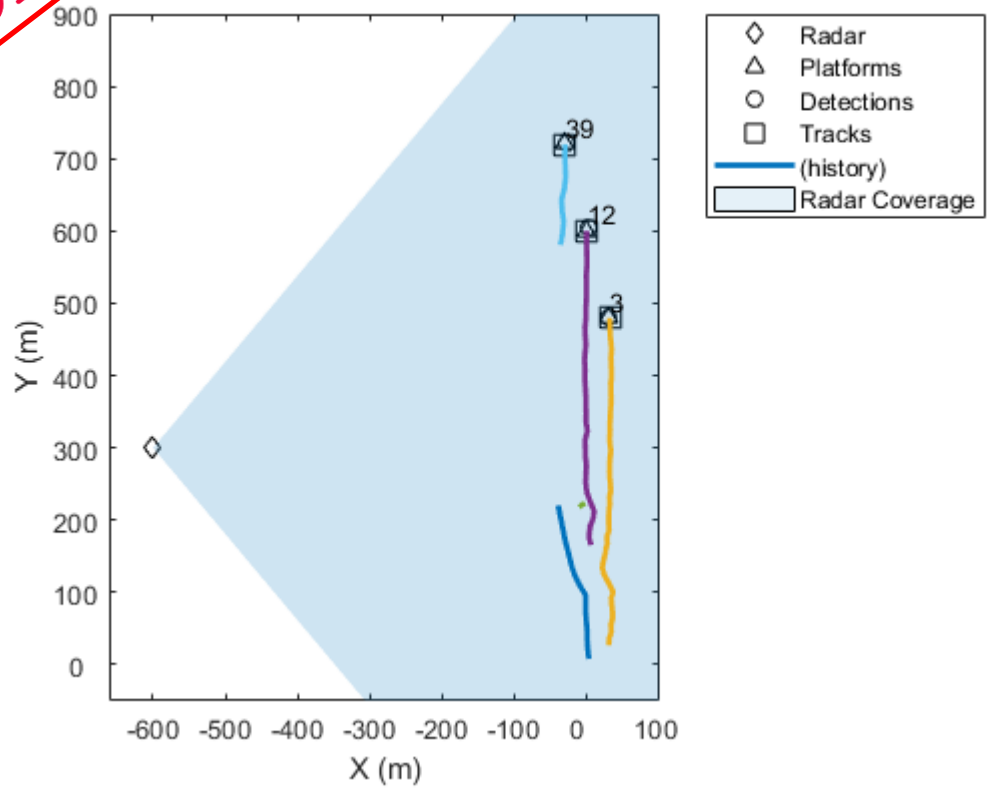


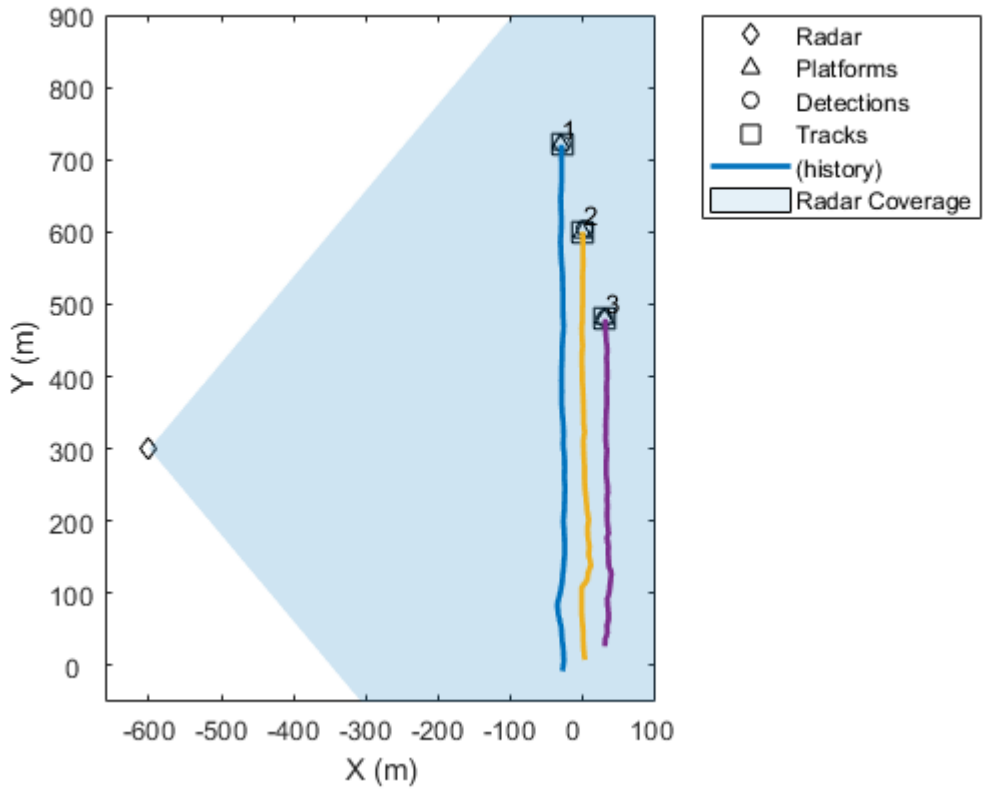
Sensor Fusion and Tracking Toolbox™

多目标跟踪器的设计和调参

DEMO

Tuning a Multi-Object Tracker



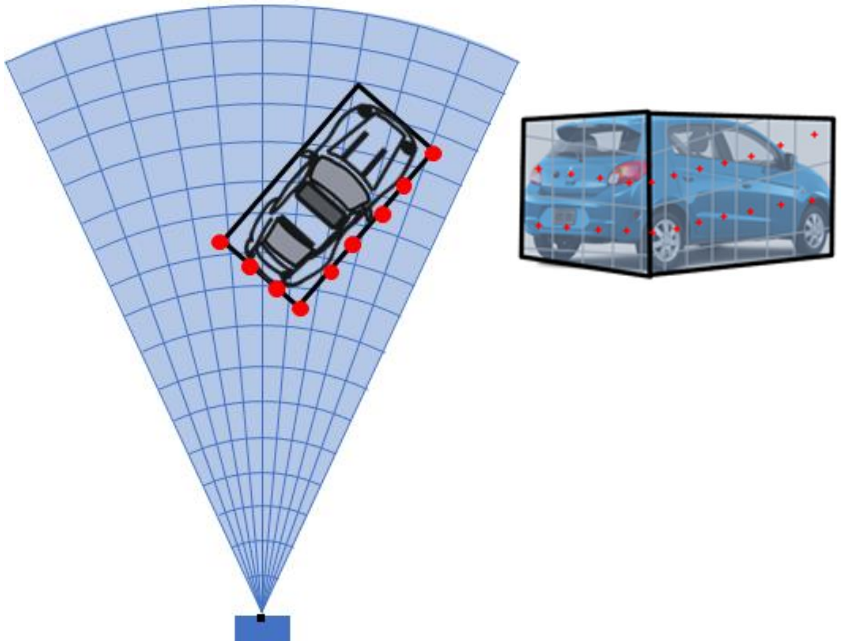


点目标 vs. 扩展目标

- **点目标**
  - 仅以一个点来表达目标
  - 单个目标生成单个检测
  - 或者以“聚类”的方法获得点目标



- **扩展目标**
  - 借助高分辨率传感器识别目标
  - 单个目标生成多个检测
  - 获得更精确的目标尺寸、方向属性



点目标跟踪与扩展目标跟踪

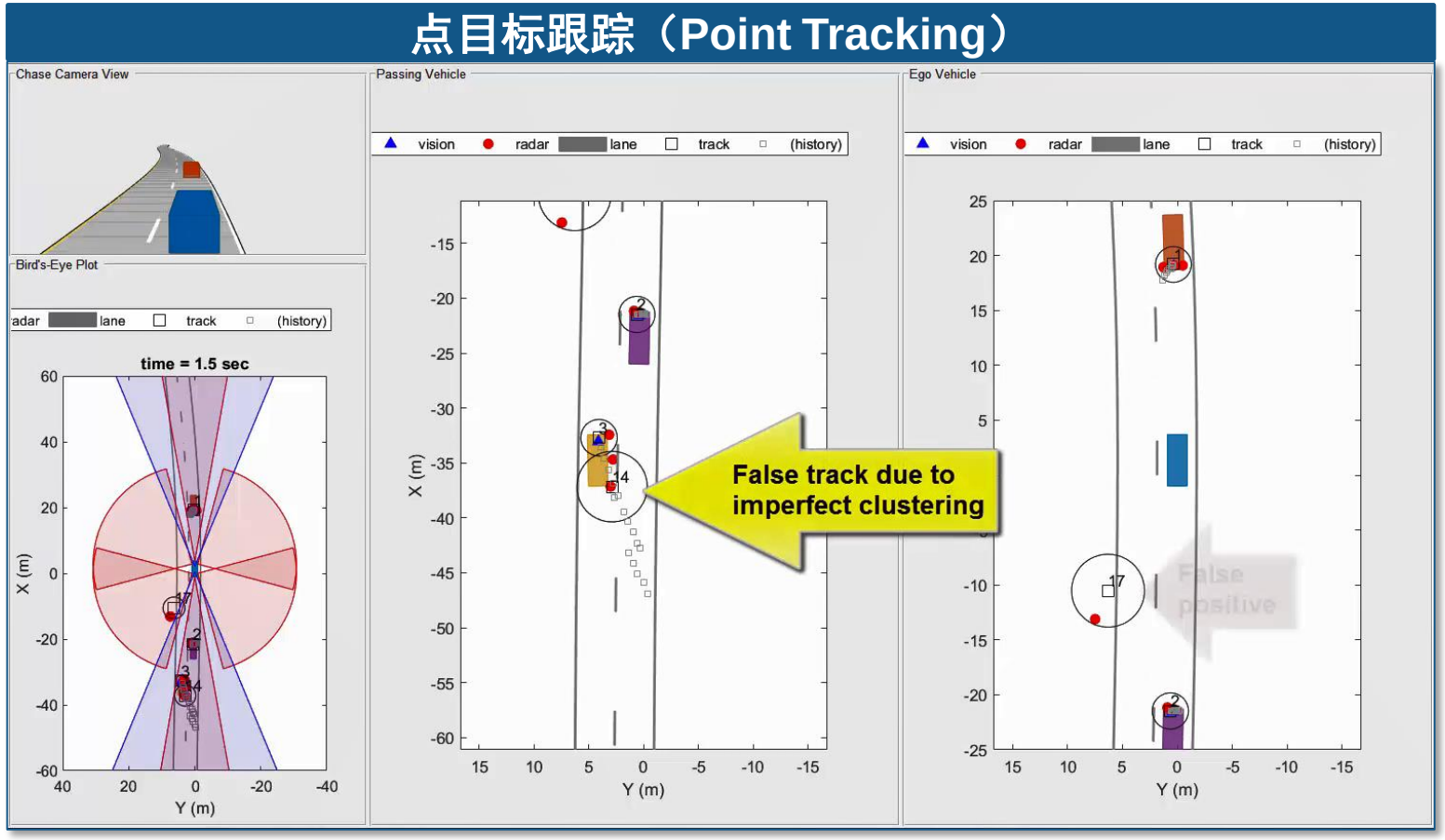
[Extended Object Tracking]

- 点目标跟踪器 multiObjectTracker
- 传感器对单个目标生成单个检测或经过聚类后形成单个检测
- 将目标简化为一个点进行跟踪

Sensor Fusion and Tracking Toolbox™

Automated Driving Toolbox™

Updated R2019b



点目标跟踪与扩展目标跟踪

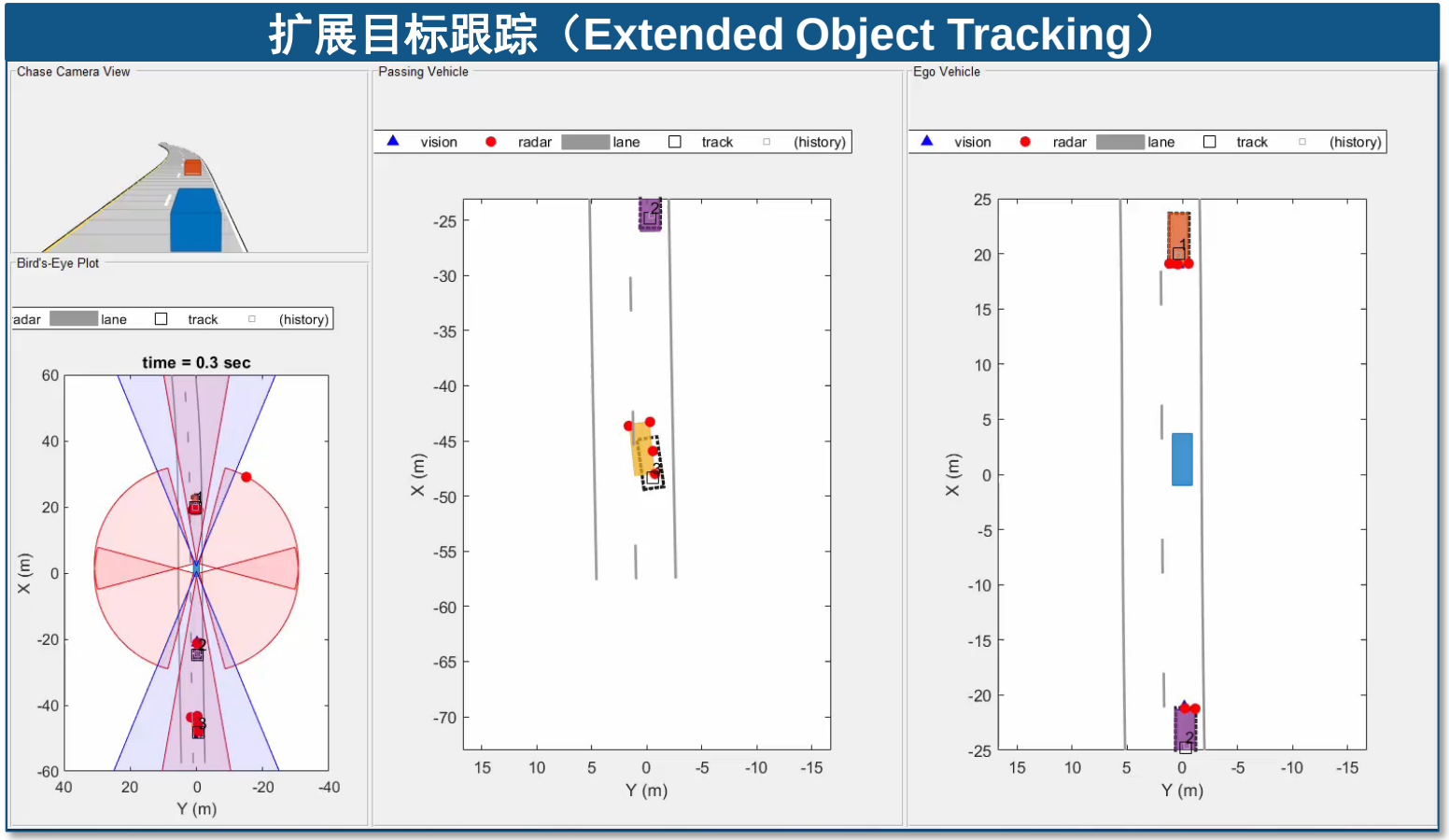
[Extended Object Tracking]

- 自定义的扩展目标跟踪器
- 利用高精度传感器对单个目标生成的多个检测
- 可获取更多目标属性：大小、形状、方向等

Sensor Fusion and Tracking Toolbox™

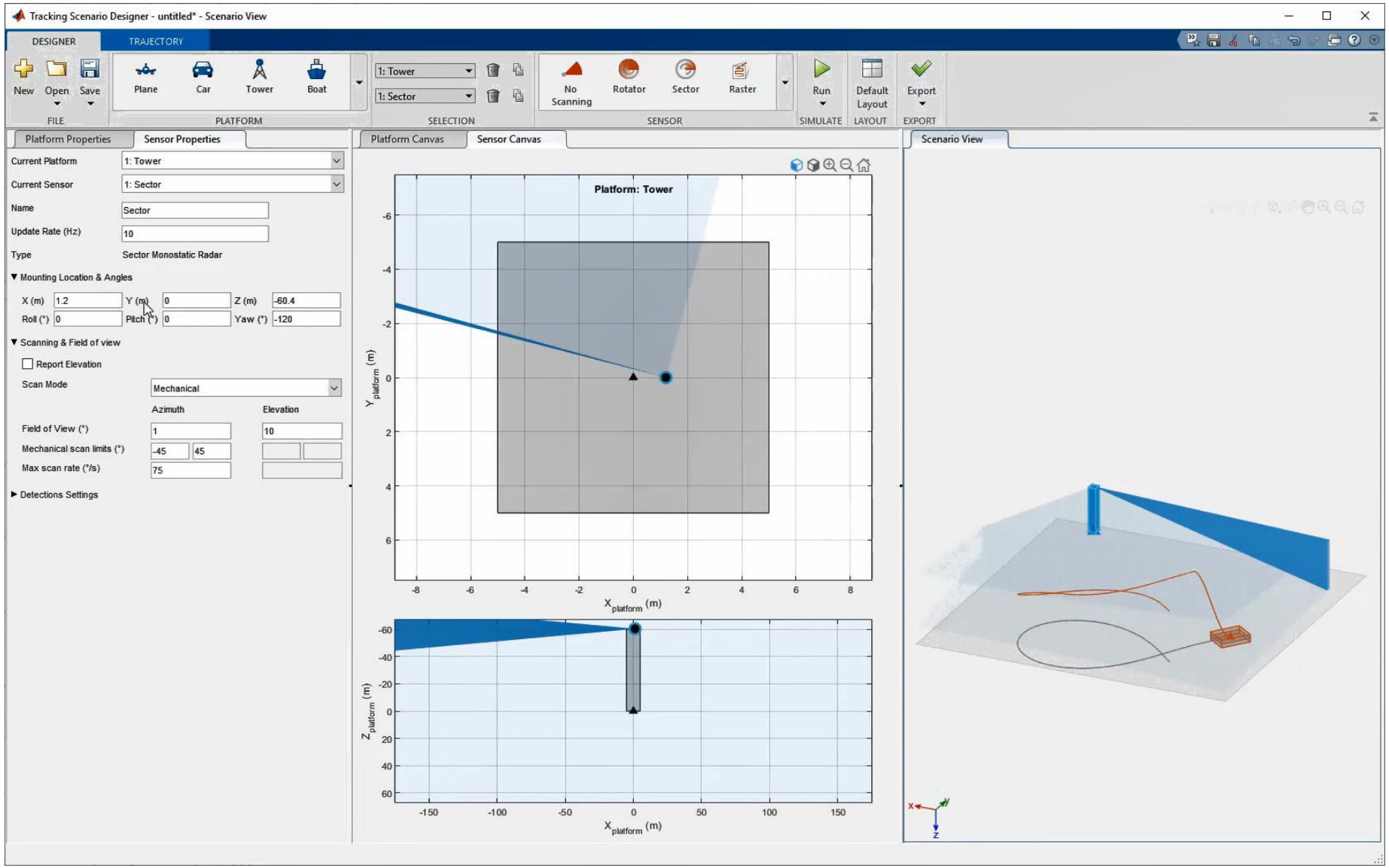
Automated Driving Toolbox™

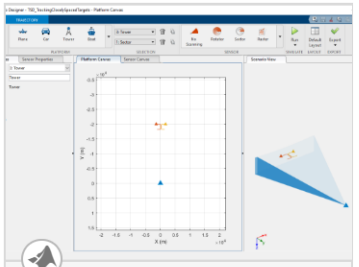
Updated R2019b



使用跟踪场景设计器创建测试场景

R2020a



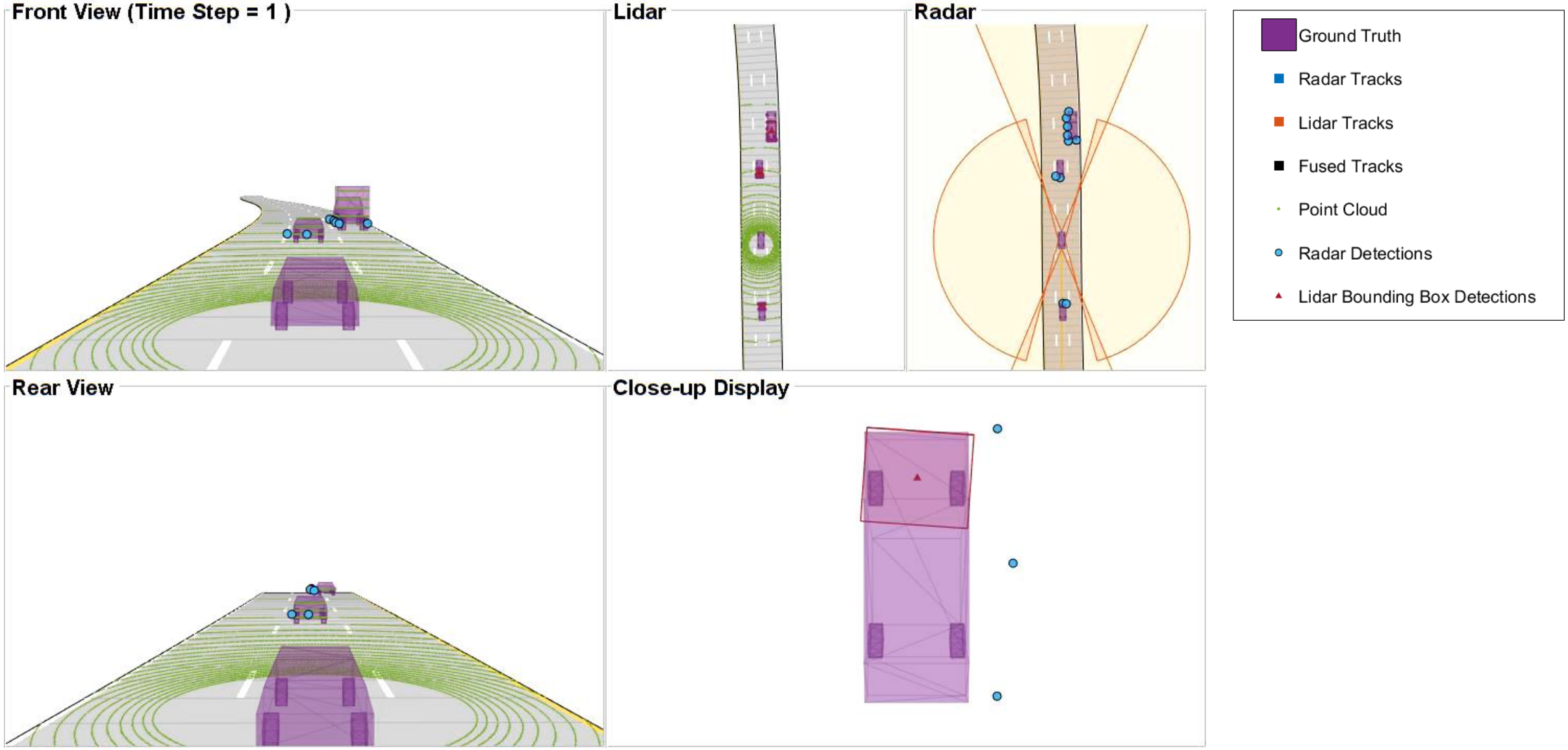


Design and Simulate a Tracking Scenario with Tracking Scenario…

Use trackingScenarioDesigner with an existing session file. Using the application, you can add, modify, or remove platforms, monostatic radar

融合激光雷达与毫米波雷达的跟踪列表

R2020a



Q&A

跟随 MATLAB 和 Simulink 教程学习

- 通过免费自定进度的在线课程，获得交互式的 MATLAB 具体操作介绍。
- https://ww2.mathworks.cn/support/learn-with-matlab-tutorials.html



2020年小迈步系列预告：

小迈步之人工智能（五）：深度学习在医学图像处理中的应用，6月30日（官网已上线）

小迈步之人工智能（六）：金融，8月27日

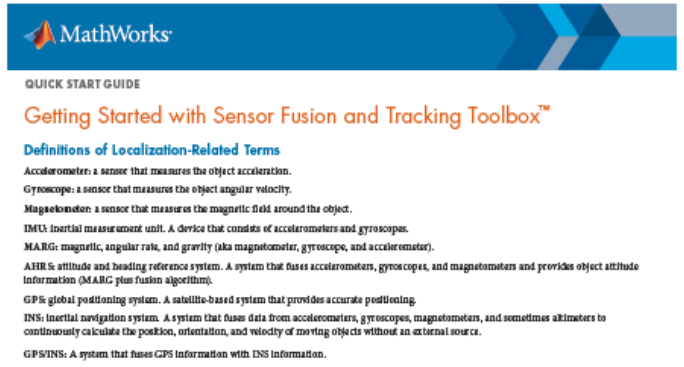
小迈步之系统仿真（一）：电力与电力电子，9月24日

小迈步之系统仿真（二）：机器人，10月22日

小迈步之系统仿真（三）：自动驾驶，11月19日

小迈步之行业规范：功能安全认证，12月10日

传感器融合快速入门（20a）


MathWorks

QUICK START GUIDE

Getting Started with Sensor Fusion and Tracking Toolbox™

Definitions of Localization-Related Terms

Accelerometer: a sensor that measures the object acceleration.

Gyroscope: a sensor that measures the object angular velocity.

Magnetometer: a sensor that measures the magnetic field around the object.

IMU: inertial measurement unit. A device that consists of accelerometers and gyroscopes.

MARG: magnetic, angular rate, and gravity (aka magnetometer, gyroscope, and accelerometer).

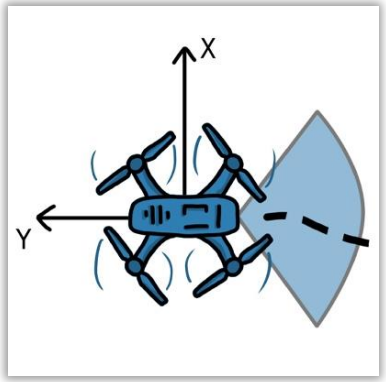
AHRS: attitude and heading reference system. A system that fuses accelerometers, gyroscopes, and magnetometers and provides object attitude information (MARG plus fusion algorithm).

GPS: global positioning system. A satellite-based system that provides accurate positioning.

INS: inertial navigation system. A system that fuses data from accelerometers, gyroscopes, magnetometers, and sometimes altimeters to continuously calculate the position, orientation, and velocity of moving objects without an external source.

GPS/INS: A system that fuses GPS information with INS information.

- 更多免费学习资源，也可以在 MATLAB 中文论坛>学习资源获取：
- http://www.ilovematlab.cn/resources/



《小迈步之人工智能（四）：实战传感器融合与目标跟踪》

填写会后调查问卷，

立刻下载小迈步课件！

上 MATLAB 中文论坛，

讨论 MATLAB 问题。

直达小迈步课程讨论帖

>> https://www.ilovematlab.cn/thread-596899-1-1.html

我们也会在后续上传今天讲的例程！

如果您有关于正版产品或服务的其他问题 请与我们联系

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